

CPSC 330 Lect

Feature Engin

and Selecti

Finishing up Feature importances and SHAP

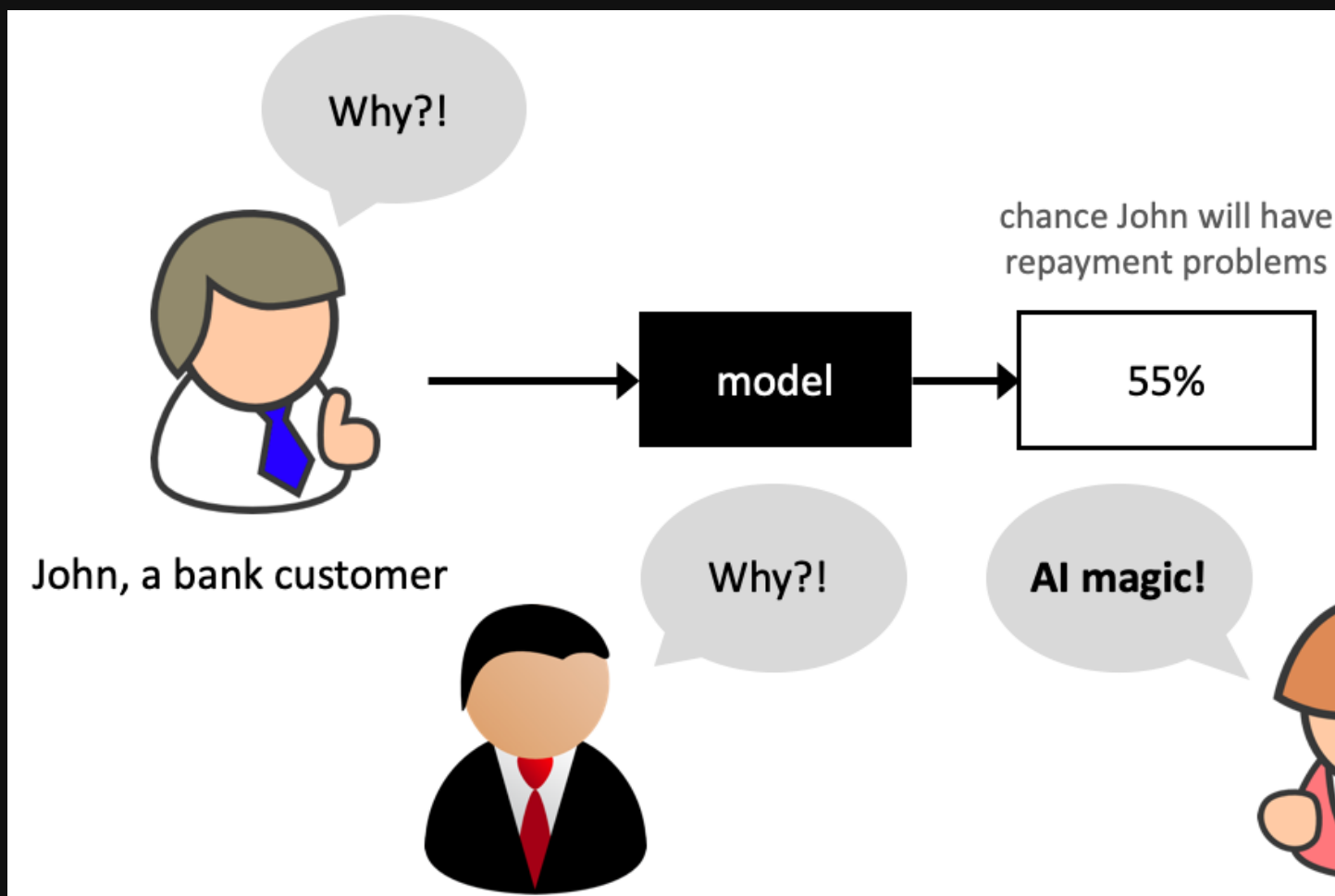
Why do we care about feature importance?

- Identify features that are not useful and remove them.
- Get guidance on what new data to collect
 - New features related to useful features.
 - Don't bother collecting useless features, saving resources.

Finishing up Feature importances

- Help explain why the model is making predictions.
 - Debugging, if the model is behaving
 - Regulatory requirements.
 - Fairness / bias. See this.
 - Keep in mind this can be used on predictions!

Why bother about model transparency?

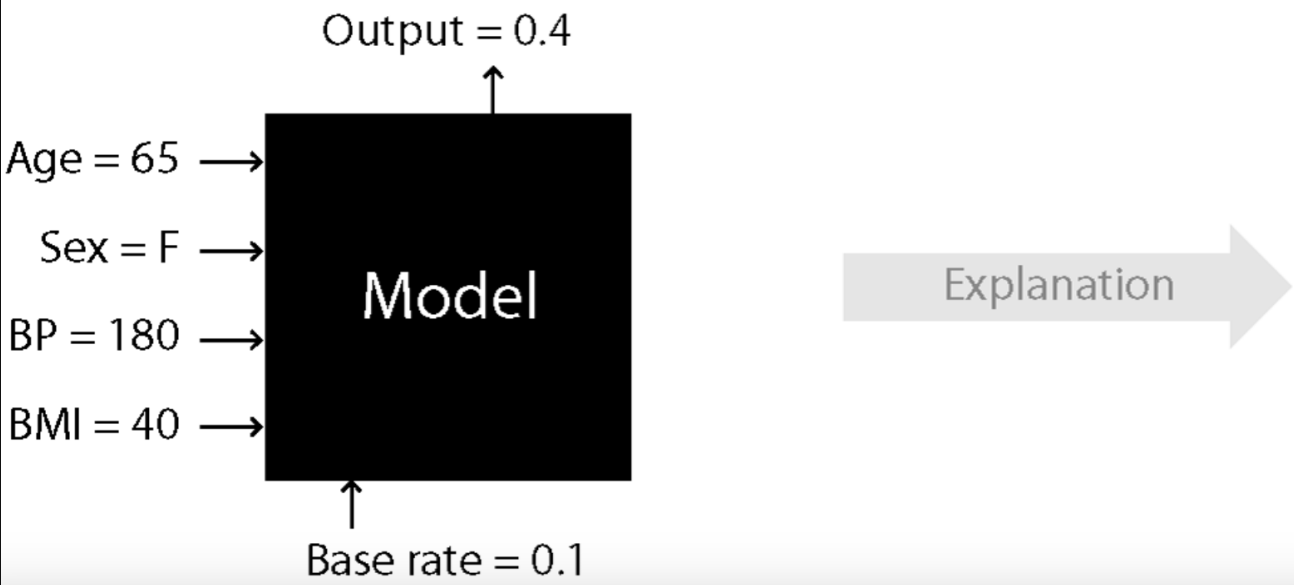


SHAP intuition

- Think of the model as a “black box” that takes inputs and produces predictions.
- SHAP asks: If we treat each feature as contributing to the final prediction, how much does each one deserve?
- To answer this fairly, SHAP looks at all possible combinations of features and averages their contributions.
- A marginal contribution is how much the prediction changes when you add that feature to a set of other features.

SHAP

SHAP



Extending SHAP

- Can also be used to explain text classification
image classification!

Extending SHAP

Source

Extending SHAP

- Example: In the picture below, red pixels represent positive SHAP values that increase the probability of the class, while blue pixels represent negative SHAP values that reduce the probability of the class.

Practice Question on SHAP

Select all the statements that are true:

- A. SHAP values are model parameters learned during training.
- B. Coefficients in a linear model and SHAP values both show how each feature contributes to a prediction, but coefficients are global and SHAP values are local.
- C. SHAP values can only be computed for tree-based models.
- D. A waterfall plot shows how each feature's SHAP value contributes to a single prediction.
- E. SHAP provides the same explanation for all examples.

Break

Let's take a break!



Clicker Question 14.0

Suppose you are working on a machine learning project. If you have to prioritize one of the following, which of the following would it be?

- A. The quality and size of the data
- B. Most recent deep neural network models
- C. Most recent optimization algorithm

Feature engine motivation

Discussion question

- Suppose we want to predict whether a flight will arrive on time or be delayed. We have the following information about flights:
 - Departure Time
 - Expected Duration of Flight (in minutes)

Upon analyzing the data, you notice that flights that depart during the middle of the day tend to be delayed more often during the winter months. What feature could be valuable for this prediction task?

Motivating Feature Eng

Questions:

- What are two possible ways we could motivate features?
 - Think broadly and philosophically about the implementation...

Garbage in, garbage out

- Model building is interesting. But in y learning projects, you'll be spending your time on data preparation, feature and transformations.
- The *quality* of the data is important. Y only as good as your data.

Activity: Measuring quality of data

- What are some properties of “good” data?
- Along what possible dimensions could we measure the goodness of data?

Dimension	Good Data	Bad Data	...

What is feature engineering

Feature engineering is the process of transforming raw data into features that better represent the underlying problem to the predictive model, resulting in improved model accuracy on unseen data.

- Jason Brownlee

- Better features: more flexibility, higher accuracy, get by with simple and more interpretable models
- If your features, i.e., representation is poor, no fancier model you build is not going to help

Some quotes on feature engineering

A quote by Pedro Domingos *A Few Useful Things That You Should Know About Machine Learning*

... At the end of the day, some machine learning projects succeed and some fail. What's the difference? Easily the most important factor is the features used.

Some quotes on feature engineering

A quote by Andrew Ng, [Machine Learning Brain simulations](#)

Coming up with features is difficult, time-consuming, and requires expert knowledge. “Applied machine learning” is basically feature engineering.

Better features usually more than a better mo

- Good features would ideally:
 - capture most important aspects of the problem
 - allow learning with few examples
 - generalize to new scenarios.
- There is a trade-off between simple and expressive fe
 - With simple features overfitting risk is low, but sco
 - With complicated features scores can be high, but

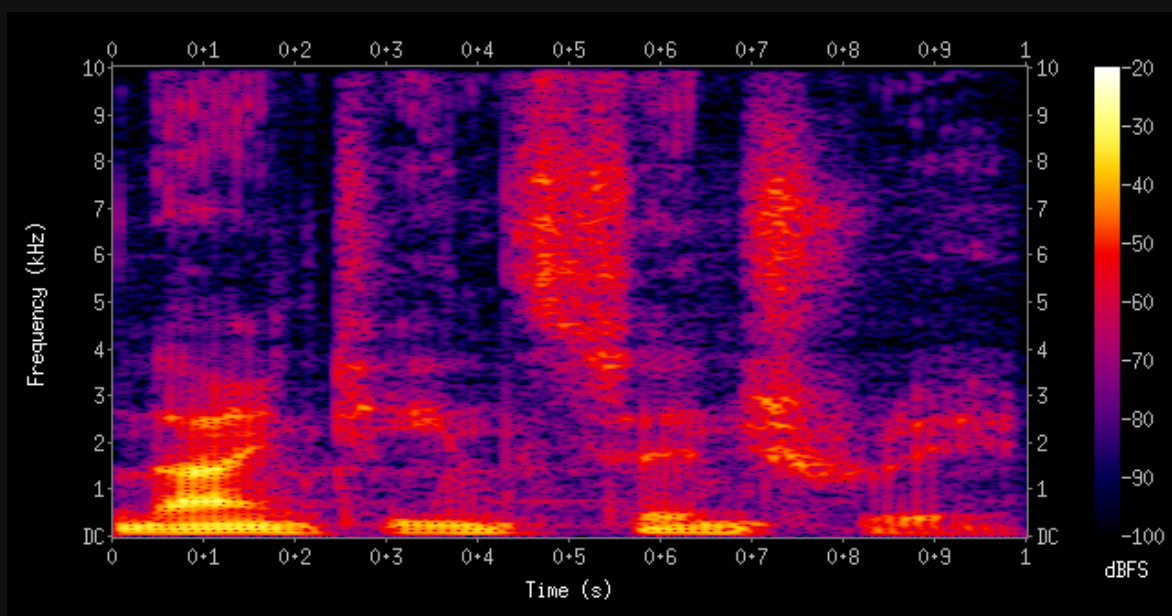
The best features may be dependent on the model you use

- Examples:
 - For counting-based methods like decision trees sensitive to the number of variable values
 - Discretization makes sense
 - For distance-based methods like KNN, we want features that are not too "far".
 - Standardization
 - For regression-based methods like linear regression, we want features that have a linear dependency on features.

Domain-specific transf

In some domains there are natural transformations that we can do:

- Spectrograms (sound data)
- Convolutions (image data)



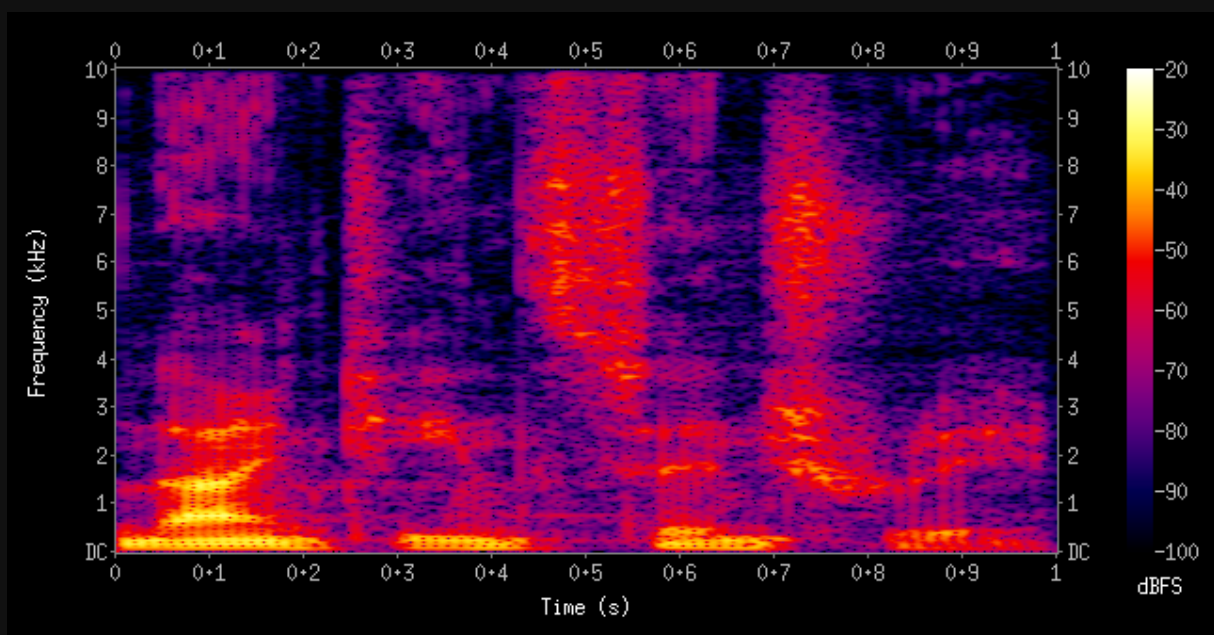
Break

Let's take a break!



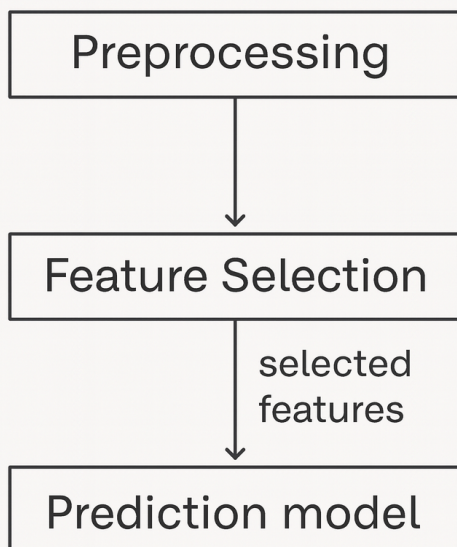
Domain-specific transf

In some domains there are natural trans-
fer do: - Spectrograms (sound data) - Wave
- Convolutions



Source

Feature selection pipeline



Two useful methods

- Model-based selection
 - Use a supervised machine learning model to estimate the importance of each feature.
 - Keep only the most important ones.
- Recursive feature elimination
 - Build a series of models
 - At each iteration, discard the least important feature according to the model.

Clicker Question 13.1

Select all of the following statements that are TRUE.

- A. Simple association-based feature selection approaches do not take into account correlations between features.
- B. You can carry out feature selection for linear models by pruning the features with small weights (i.e., coefficients less than a threshold).
- C. The order of features removed given by `rfe.ranking_` is the same as the order of features removed by `rfe.feature_names_`.

(iClicker) Exercise 13.2

Select all of the following statements that are TRUE.

- A. You can carry out feature selection on linear models by pruning the features with small weights (i.e., coefficients less than a threshold).
- B. The order of features removed given by `rfe.ranking_` is the same as the order of feature importances given by the model.

Group Work: Class Demo

Coding (if time permits)

For this demo, each student should [click](#) create a new repo in their accounts, then clone the repo locally to follow along with the demo.